

0003 — Keystone: The Initial-Balance Edge Fade — 2026-06-24

****Series:**** MC Setup Research Notes · Note 0003 ****Confidence:**** Medium — 5 years MC, full-cost, PASSED a look-ahead audit, and positive in every calendar year (a quasi-out-of-sample check). Held back from High because it is still in-sample on MC only, was selected among ~85 tested filters (so the forward edge is the conservative end of the range), and carries a deep drawdown.

TL;DR: We searched for whether an MC signal's location — where its channel originated relative to structural levels — predicts a tradeable edge. Almost everything failed (reversal-at-extreme, prior-day levels, volume nodes, single prints, value-area edges, an IB-width "sweet spot" that turned out to be a stop-size illusion). **One survived: signals whose origin sits within 0.10 ADR of the day's Initial Balance (first-hour) edge — a responsive fade back into value.** At a fixed 2.0R exit it makes **+0.159 R/trade (PF 1.38, 1,395 trades, ~\$233k on one contract over 5yr)**, the filter concentrates the edge (the rest of the book is flat), it works long and short, and it survives realistic costs and a look-ahead audit. **But it is a thin, deep-drawdown edge** (~\$26k max DD/contract; realistic forward ~+0.09–0.12R after costs and selection bias). **Recommendation: a sized component for a well-capitalized cash account — not a standalone automated prop system.**

1. The setup (so this note stands alone)

The MC CC trade. The "MC" indicator fires breakout signals (subtypes CC1–CC5) on 5-minute ES. You enter at the first tick after the signal bar's close, place a stop at the channel's protective level (the MCX extreme), and exit at a fixed multiple of **R** (R = entry-to-stop distance). ES: 1 point = \$50/contract, 1 tick = \$12.50.

Initial Balance (IB). The IB is the **high/low of the first 60 minutes** of the RTH session — the auction's opening agreed-upon range. In Auction Market Theory, a move that launches off the IB boundary and rotates back in is a **responsive** trade: the market tested the early extreme and rejected it.

Keystone, in one sentence. Keep only those MC signals whose **channel origin** (the StopPrice/MCX extreme — the swing the channel formed from) sits **within 0.10 ADR of the same-side IB edge**: longs whose origin is at the IB low, shorts whose origin is at the IB high. Then trade them with a plain fixed **2.0R** target, single contract, both directions.

Distance metric. $d = |\text{origin} - \text{IB edge}| / \text{ADR}$, where ADR = prior-day ATR(14) in points. The gate is $0 \leq d \leq 0.10$. Normalising by ADR makes the threshold regime-independent.

Cost model. \$4.36 round-turn commission + 1 tick entry slippage, 0 exit (the MC pin; slightly lighter than the house default of \$5 + 1 tick/leg — immaterial, and we stress heavier costs in §4.7). All sims replay **real continuous ticks day-by-day**. Signal set: `ba_signals_mc.parquet` (5,580 signals, 2021-06 → 2026-06). Features are tagged **causally** through the S34-corrected `tag_signals` chokepoint (see Note 0002 for the look-ahead fix this depends on).

2. The question

1. **Does an MC signal's origin location, relative to a structural level, select winners?** We pre-committed a menu of level families rather than fishing one bucket at a time.
2. **For the one family that worked (the IB edge): can we make it better** — a different threshold, exit, management rule, leg structure, direction, or filter stack — or is the simple version the best version?
3. **Is the surviving edge real, or another look-ahead / overfit artifact?**

3. How we tested it (the menu)

Finding it — location families (each = a descriptive bucket study on the real tick engine, pinned 1.0R single-leg, $\pm 95\%$ CIs, by-year):

1. Reversal off the **developing day extreme** (LOD/HOD).
2. Origin at the **prior-day high/low** (LOY/HOY), incl. conditioned on a balance day.
3. **Balance state** (opened inside yesterday's range and still rotating).
4. **Failed-breakout fade** ("look above/below and fail").
5. Origin at prior **volume nodes** (HVN / LVN) and **single/zero prints** — built a profile-node extractor for this.
6. Origin at the prior **value-area edge** (VAH / VAL).
7. **IB width "sweet spot"** (is rotation cleanest at a mid IB width?).
8. Origin at the **IB edge** — the survivor.

Improving it (on the IB-edge book): a 0.05-ADR **threshold sweep** (where to draw the gate line); a **target sweep** 0.5R–4R; **active management** (breakeven, lock-in trail); **2-leg scale-in / scale-out**; **direction split**; **filter stacking** with balance.

Breaking it (the audit): a **session-timing split** (does the edge live only where the IB is still forming?), **cost stress** to 3-tick slippage, and a **StopPrice causality** sanity check.

Engine: tick-by-tick first-touch (entry at first tick after the signal; stop/target by tick crossing). Memory-frugal 4-chunk tick load for the full-population sims.

4. Results

4.1 The location search — Keystone was the lone survivor

Hypothesis (origin / location vs a level)	Verdict
Reversal off developing day low/high (LOD/HOD)	REJECT — long-only, drift-suspected
Origin at prior-day low/high (LOY/HOY)	REJECT — nothing beyond "balance"
Balance state (opened inside Y, still rotating)	WEAK — real but fails 2023–24
Failed-breakout "look above/below & fail" fade	REJECT — the canonical fade LOSES
Origin at prior volume nodes (HVN / LVN)	REJECT — null
Origin at single / zero prints (profile gaps)	REJECT — null
Origin at prior value-area edge (VAH / VAL)	WEAK — suggestive but thin (n≈100)
IB width "sweet spot" (inverted-U in dollars)	REJECT — a stop-size illusion in \$
Origin at the IB edge (≤ 0.10 ADR) = KEYSTONE	SURVIVED everything below

Across ~85 buckets, almost everything died honestly. Reporting the failures is the point: it shows the survivor was not cherry-picked from one lucky look. (The IB-width case is instructive — it looked great in dollars because wider IB days have wider stops, i.e. more \$ per winning R; in R the apparent peak vanished. We judge everything in R for exactly this reason.)

4.2 Headline performance (1 contract, ~5 yr, 2.0R)

Trades	Net P&L	Exp R	Profit Factor	Win %	Net / MaxDD
1,395	\$232,593	+0.159	1.38	48.5%	8.86

716 long / 679 short; ~280 trades/year (about one per session).

4.3 The filter does real work — selection value at 2.0R

Population	Trades	Exp R	Profit Factor
Keystone gate	1,395	+0.159	1.38
Everything else (non-gate)	4,049	+0.026	1.08
All signals (baseline)	5,444	+0.060	1.17

Same 2.0R exit for all three rows. **The non-gated remainder is essentially flat** — the gate concentrates the edge rather than slicing an already-good book. That clean separation (gate vs the inert rest) is the core evidence the filter carries real information.

4.4 Where to draw the gate line — a CLIFF at 0.10 ADR, not a knob

Sweeping origin-to-IB-edge distance in 0.05-ADR bands (full population, 1.0R basis):

Distance to IB edge (ADR)	Exp R	Profit Factor
0.00 – 0.05	+0.097	1.25
0.05 – 0.10 (gate keeps ≤ 0.10)	+0.156	1.51
0.10 – 0.20	-0.02	0.99
0.20 – 0.35	+0.04	1.13

The edge is concentrated within 0.10 ADR and **falls off a cliff** beyond it. 0.10 is a structural boundary, not a fitted peak — there is no smooth knob to tune. Tighter (0.05) throws away half the trades for no better R.

4.5 The exit target — a plateau, not a tuned peak

Target	Exp R	Profit Factor	Net / MaxDD
1.0R	+0.112	1.30	6.9
1.5R	+0.136	1.34	7.8
2.0R (chosen)	+0.159	1.38	8.9
3.0R	+0.161	1.35	8.3
4.0R	+0.169	1.37	9.1

Expectancy rises and then **plateaus from ~2R**. 2.0R sits on a flat shelf — a tuned peak would tower over its neighbours; this doesn't. We take 2.0R as the conservative entry to the plateau.

4.6 Improving it further — nothing beat the simple version

Active management (gated book, 2.0R):

Exit rule	Exp R	Net / MaxDD
Plain 2.0R (chosen)	+0.159	8.9
+ break-even after +1R	+0.161	9.4
+ trail (lock +0.5R after +1.5R)	+0.158	8.5

Break-even is a hair better on MAR but trades win-rate for it and adds moving parts — the gain is inside the noise.

Two-leg scale-in / scale-out. Adding a second contract on a pullback (including the "E1 scratches, E2 wins" structure) and scaling a partial out at 1R **both underperformed plain single-leg 2.0R on net AND drawdown** — the pullback-add piles size into the losers, deepening the DD. (This work also uncovered and fixed a P&L mis-scaling bug in the 2-leg engine; see BUG_multileg_pnl_scaling.md. Keystone is single-leg and unaffected.)

Direction. Long-only +0.091R, short-only +0.133R, **both positive in every year**. We trade both — the symmetry is itself

evidence the edge is structural; dropping a side to flatter the curve would be fitting.

Stacking with balance. Layering the "balance day" context lifts the gate to +0.168R — but balance alone is weak and regime-dependent, and it halves the trade count. We keep the gate **standalone** (more trades, cleaner); balance is an optional conviction add, not part of the core rule.

The pattern across all of §4.6: **the edge lives in selection (which trade to take), not in management after entry.** The simplest config is the best config.

4.7 Consistency, costs, and the look-ahead audit

By year (Exp R, 2.0R):

2021	2022	2023	2024	2025	2026
+0.315	+0.207	+0.034	+0.121	+0.198	+0.174

Positive every year, including the 2022 bear. 2023 is the softest (thin positive); no single year carries the result.

Execution cost stress (Exp R, 2.0R):

Slippage assumption	Exp R	Profit Factor
1 tick in / 0 out (base)	+0.159	1.38
2 in / 1 out (realistic-conservative)	+0.123	1.30
3 in / 2 out (brutal)	+0.092	1.24

Stays clearly positive even under pessimistic fills (lower 95% CI bound above zero at brutal).

Look-ahead audit — the decisive test (PASSED). If the edge came from peeking at the final IB before it formed, it would concentrate in signals fired DURING the first hour (IB still developing) and vanish AFTER it. The opposite holds:

Subset	Trades	Exp R	Profit Factor
After first hour (IB fully PAST)	603	+0.203	1.62
During first hour (IB developing)	792	+0.126	1.27

The edge is **stronger** where the IB is indisputably in the past. OR60 is causal in code (developing running range during the first hour, frozen after — indicators.py:263-274); no as-of merge can land on the entry bar; StopPrice is 100% on the correct side of the signal. The signature of a real effect, not a leak.

5. Why it works (and why it's thin)

Why it works. The IB edge is a level the whole session references. An MC channel that originates there — i.e. forms off the first-hour high or low — is a structurally-located responsive trade: the market reached the early boundary, was rejected, and the breakout carries it back toward developing value. That the effect is symmetric (both sides), every-year-positive, and stronger once the IB is fully formed all point to a genuine auction mechanic rather than a data artifact.

Why it's thin. It is a selection edge on a base population that, corrected for look-ahead, has almost no standalone edge (Note 0002). A filter can only concentrate existing structure, not manufacture it — so the ceiling is modest. After realistic costs and the winner's-curse haircut from testing ~85 buckets, the honest forward number is the **+0.09–0.12R** end, not +0.159R. And the drawdown is deep and **regime-clustered** (the losers arrive in runs during weak years), which no exit choice repairs — it is intrinsic to what the gate selects.

6. Recommendation

1. Keystone is a real, audited, modest edge — deploy it only as a sized COMPONENT. Single-leg, 2.0R, both directions, gate at 0.10 ADR, no per-period optimization. It is **not** a standalone automated prop system: on one contract the worst drawdown was ~\$26k and the book sits >\$5k below its high-water mark ~48% of the time, with a ~9-month recovery. That path will trip a tight trailing-DD prop account. It suits a **well-capitalized cash account** (~\$75–100k of capital per ES contract) or a **sizing/selection overlay** on a broader book where its bleeds dilute.

2. Keep it simple — do not add management. Every improvement we tried (break-even, trail, scale-in, scale-out, direction-pruning, filter-stacking) failed to beat the plain version risk-adjusted. The edge is in selection; management only adds overfit surface.

3. Before committing capital: confirm out-of-sample on truly held-out data (and on RevFT), then run the prop/cash account simulator with contract scaling and the never-blow floor for blow-up probability and net-to-trader — **the drawdown path, not the average, is the binding constraint.**

7. Caveats & open questions

- **In-sample, MC only.** All figures are over the full 2021–2026 MC history. The by-year positivity is a quasi-OOS reassurance, not a true held-out test. Not yet replicated on RevFT.

- **Selection bias.** Found among ~85 tested buckets; the look-ahead audit clears leakage but not multiple-testing — hence the conservative forward read.
- **StopPrice residual.** The origin comes from the NT MC indicator export; its internal causality can't be audited from the Python side. Low risk (it's the same stop every MC strategy uses, and it is 100% on the correct side of the signal), but not zero.
- **Deep, regime-clustered drawdown** is the real obstacle, and it is **target-invariant** — 1R through 4R all carry ~\$25k max DD. This is an entry/selection property, not an exit one.
- **Costs** here are slightly lighter than the house default; §4.7 shows the edge survives to 3-tick slippage, so conclusions are not cost-sensitive.

8. Reproduce

Scripts (headless, day-by-day tick replay against the production engine; no production code changed except the documented 2-leg P&L fix):

- scripts/origin_at_extreme_study.py — LOD/HOD & LOY/HOY reversal buckets (§4.1)
- scripts/balance_refine_study.py — balance state, IB / gap / value-area conditioning (§4.1, §4.6)
- scripts/profile_node_study.py — HVN/LVN/single-print origin (§4.1)
- scripts/ib_study.py — IB width inverted-U + the IB-edge distance gradient (§4.1, §4.4)
- scripts/ib_edge_deep.py — gate vs complement, symmetry, year-by-year, cost, redundancy (§4.3, §4.6, §4.7)
- scripts/ib_edge_target_sweep.py — target sweep / plateau (§4.5)
- scripts/ib_edge_exits.py — break-even / trail / scale-in / scale-out (§4.6)
- scripts/ib_edge_scalein.py — the "E1 scratch / E2 win" scale-in + 2-leg accounting check (§4.6)
- scripts/ib_edge_equity.py — stitched equity & drawdown path (§5, §6)
- scripts/keystone_audit.py — session-timing split, cost stress, StopPrice sanity (§4.7)

Saved artifacts (in docs/living/): origin_at_extreme_study_20260623.md, balance_refine_study_20260623.md, profile_node_study_20260623.md, ib_study_20260623.md, ib_edge_deep_20260623.md, ib_edge_target_sweep_20260623.md, ib_edge_exits_20260624.md, ib_edge_scalein_20260624.md, ib_edge_equity_20260623.md, keystone_audit_20260624.md.

Gate definition (look-ahead-safe): $d = (\text{StopPrice} - \text{OR60_Low})/\text{ADR}$ for longs, $(\text{OR60_High} - \text{StopPrice})/\text{ADR}$ for shorts; keep $0 \leq d \leq 0.10$. ADR = prior_ATR; OR60/ADR via the S34-fixed tag_signals.